

Alexandre Perozim de Faveri

CONTACT INFORMATION	EPFL SB MATH CM 1 617 (Centre Midi) Station 10, 1015 Lausanne Switzerland	alexandre.defaveri@epfl.ch https://afaveri.github.io
RESEARCH INTERESTS	Analytic number theory and its interactions with harmonic analysis, automorphic forms, dynamical systems, spectral theory, and additive combinatorics.	
EMPLOYMENT	École Polytechnique Fédérale de Lausanne (EPFL) Bernoulli Instructor	2025 – Present
	Stanford University Postdoctoral Scholar	2023 – 2025
	Institute for Advanced Study (IAS) Member	2022 – 2023
EDUCATION	California Institute of Technology Ph.D. in Mathematics Advisor: Maksym Radziwiłł	2018 – 2022
	Princeton University A.B. in Mathematics Advisor: Peter Sarnak	2014 – 2018
PREPRINTS AND PUBLICATIONS	8. Non-orthogonality of the cubic and quartic large sieves via Rankin–Selberg Joint with A. Dunn and J. Hoffstein <i>Submitted</i> (2026), 41 pp. 7. Non-vanishing for quartic Hecke L-functions and ranks of elliptic curves Joint with C. Castillo and A. Dunn <i>Submitted</i> (2026), 65 pp. 6. Mass equidistribution for lifts on hyperbolic 4-manifolds Joint with Z. Shem-Tov <i>Submitted</i> (2026), 31 pp. 5. Non-vanishing for cubic Hecke L-functions Joint with C. David, A. Dunn, and J. Stucky <i>Submitted</i> (2024), 92 pp. 4. Non-escape of mass for arithmetic quantum limits on hyperbolic 4-manifolds Joint with Z. Shem-Tov <i>Submitted</i> (2024), 29 pp. 3. Simple zeros of $GL(2)$ L-functions <i>Journal of the European Mathematical Society (JEMS)</i> 27 (2025), no. 7, 2985–3021 2. The variance of closed geodesics in balls and annuli on the modular surface <i>Advances in Mathematics</i> 403 (2022), Paper No. 108390, 45 pp. 1. Möbius disjointness for $C^{1+\varepsilon}$ skew products <i>International Mathematics Research Notices. IMRN</i> (2022), no. 4, 2513–2531	

AWARDS AND FELLOWSHIPS	1. Scott Russell Johnson Dissertation Prize (Caltech Math Department) for the best graduate dissertation in mathematics	2022
	2. Scott Russell Johnson First-Year Prize (Caltech Math Department) for excellence in first-year graduate studies	2019
	3. Phi Beta Kappa	2018
	4. Andrew H. Brown Prize (Princeton Math Department) for outstanding juniors in mathematics	2017
	5. Shapiro Prize for Academic Excellence (Princeton University)	2016
	6. Fellow of Fundação Estudar	2014
	7. IMO Bronze Medal	2014
TEACHING	Instructor at EPFL	
	• Math 437	Calculus of Variations
	• Math 202(c)	Analysis III (Vector Calculus for Engineering)
	• Math 327	Topics in Complex Analysis
		Spring 2027
		Autumn 2026
		Autumn 2025
	Instructor at Stanford	
	• Math 20	Calculus (Head Instructor)
	• Math 152	Elementary Theory of Numbers
	• Math 110	Number Theory for Cryptography
		Spring 2025
		Winter 2024
		Autumn 2023
	Teaching Assistant at Caltech	
	• Math 3	Probability and Statistics
	• Math 17	Putnam Seminar (Instructor)
	• Math 11	Mathematical Writing
	• Math 17	Putnam Seminar (Instructor)
	• Math 3	Probability and Statistics
	• Math 110	Graduate Real Analysis
	• Math 3	Probability and Statistics
	• Math 2	Differential Equations
		Winter 2022
		Fall 2021
		Spring 2021
		Fall 2020
		Winter 2020
		Fall 2019
		Winter 2019
		Fall 2018
	Teaching Assistant/Grader at Princeton	
	• MAT 218	Accelerated Honors Analysis
	• MAT 325	Complex Analysis
	• COS 423	Theory of Algorithms
	• COS 226	Algorithms and Data Structures
	• COS 226	Algorithms and Data Structures
	• COS 226	Algorithms and Data Structures
		Spring 2018
		Fall 2017
		Spring 2017
		Fall 2016
		Spring 2016
		Fall 2015
SERVICE	Students Supervised	
	□ Jan Michal Radominski-Lasek (Bachelor's thesis, École polytechnique)	2026
	Seminars Organized	
	□ Organizer of the Stanford Analytic Number Theory Seminar	2023 – 2025
	□ Organizer of the Caltech Graduate Student Seminar	2019 – 2020
	Journal Refereeing	
	□ Quick opinions: Annals of Mathematics • Journal of the American Mathematical Society • Journal d'Analyse Mathématique	

- ❑ Referee: Forum of Mathematics. Pi • Communications in Mathematical Physics • International Mathematics Research Notices. IMRN • Journal of the London Mathematical Society • Mathematische Zeitschrift • Journal de Théorie des Nombres de Bordeaux • Proceedings of the American Mathematical Society • Acta Arithmetica • Journal of Number Theory • Forum Mathematicum • Canadian Journal of Mathematics • Bulletin of the Brazilian Mathematical Society • Bulletin of the Australian Mathematical Society • Canadian Mathematical Bulletin
- ❑ Reviewer: Mathematical Reviews

Other

- ❑ Lecturer for the Summer School on Analytic Number Theory 2025 (Xi'an, China)
- ❑ External expert for the Master's thesis of Erasmo Santini (EPFL, 2026)
- ❑ Editor of the AIM Problem List on Sarnak's Conjecture
- ❑ Problem Coordinator for IMO 2017 (Rio de Janeiro, Brazil)
- ❑ Grader for CHMMC and PUMaC

RESEARCH VISITS

1. Centre de Recherches Mathématiques (CRM), April to July 2026
2. Sorbonne Université, April 2026
3. Georgia Tech, November 2024
4. Concordia University, October 2024
5. IAS, October 2024
6. UT Austin, April 2022
7. University of Bristol, November 2021
8. UT Austin, October 2021

INVITED TALKS

1. *Kansas State Number Theory Seminar*
Kansas State University (online), October 2026
2. *The Fourth JNT Biennial Conference*
Cetraro, July 2026
3. *Probability in Number Theory Seminar*
Université de Montréal, June 2026
4. *Recent Perspectives on Moments of L-Functions*
UT Dallas, June 2026
5. *ETIENE Joint Meeting, Bordeaux*
Université de Bordeaux, April 2026
6. *University of Virginia Number Theory Seminar*
University of Virginia (online), November 2025
7. *AMS Spring Eastern Sectional Meeting*
Hartford, April 2025
8. *CRG Weekly Seminar Series*
Online, January 2025
9. *Copenhagen Number Theory Seminar*
University of Copenhagen (online), December 2024
10. *CDSNS Colloquium*
Georgia Tech, November 2024
11. *Georgia Tech Number Theory Seminar*
Georgia Tech, November 2024

12. *UC Davis Algebraic Geometry and Number Theory Seminar*
UC Davis, November 2024
13. *Quebec-Vermont Number Theory Seminar*
Concordia University, October 2024
14. *Southern California Number Theory Day*
UC Irvine, October 2024
15. *Purdue Analytic Number Theory and Harmonic Analysis Seminar*
Purdue University, October 2024
16. *Joint IAS/PU Groups and Dynamics Seminar*
Institute for Advanced Study, October 2024
17. *Shanks Conference Series*
Vanderbilt University, May 2024
18. *Stanford Number Theory Seminar*
Stanford University, April 2024
19. *FRG Grad Seminar*
Online, February 2024
20. *Around Frobenius distributions and related topics IV*
Online, October 2023
21. *IAS Members' Colloquium*
Institute for Advanced Study, April 2023
22. *Crosscurrents in number theory, analysis and geometry*
Université Sorbonne Paris Nord (online), December 2022
23. *UCLA Number Theory Seminar*
University of California, Los Angeles, April 2022
24. *First International Workshop in ANT at UNIST*
Ulsan National Institute of Science and Technology (online), April 2022
25. *Heilbronn Number Theory Seminar*
University of Bristol, November 2021
26. *Number Theory Seminar*
University of Mississippi (online), October 2021

PERSONAL

Age: 30
Citizenship: Brazilian